A GAME-CHANGER CALLED CERVAVAC

On September 2, the Serum Institute of India in Pune launched Cervavac, the country's first indigenously developed and affordable cervical cancer vaccine. Priced at Rs 200-400 per dose, it is significantly more accessible than the options currently available in India: quadrivalent vaccine Gardasil (Rs 2,800 per dose) and bivalent vaccine Cervarix (Rs 3,299 per dose). The government already has plans to buy 10 million doses by the year-end.

Cervavac holds out hope for millions of women at risk in India. The country accounts for about a fifth of the global burden of cervical cancer, recording about 123,000 cases and around 67,000 deaths every year, according to World Health Organization data. But almost 80-90 per cent of the cases can be prevented by a quadrivalent HPV vaccine. Cervavac, like the hepatitis B vaccine, is based on virus-like particles or VLPs—and provides protection by generating antibodies against the HPV virus's L1 protein. It has reportedly shown a robust antibody response, nearly 1,000 times higher than the baseline against all HPV strains and across age groups. Its affordability makes it an even more potent weapon against cervical cancer.

"The new immune therapies, either to detect cancer early or to modify a cell, are personalised, differing from case to case as everyone has unique genes and cells"



DR ASHOK VAID
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product is foreign) to be produced in large numbers. "Anything in healthcare is always expensive when it starts—be it any technology or robotic surgery," he says. "But, gradually, technology changes and scaling up happens and the dynamics of cost structure undergo dramatic change; it is a natural progression for costs to reduce over time. To fasten the pace of reduction, having a financial intermediary, either public or private, will also help." Developing CAR-T products India should also bring down the final price to some extent, he feels.

Demand, too, should ease expense. "We do about 14 per cent of all heart surgeries in India," he says. "Sometimes, we have 40 surgeries going on in just one building in a day. With that volume, your outcome improves because you have surgeons with great skills, and with the economies of scale, the cost of everything comes down." However, for more people to opt for these new technologies, misinformation and the lack of awareness, experts say, need to be tackled first.

Currently, awareness of new treatment options is limited to a few pockets in the country. Experts advocate more active engagement among patients, physicians and scientists so that the general public gets to know about the new advances in diagnostics and treatments. Many people also fear anything that involves 'genetic modification', scared that altering the human gene could change other things in a person. All these treatments, however, go through extensive long-term safety trials. Purwar's CAR-T therapy, for example, uses lentiviral vectors (a type of retrovirus that can get through the nucleic membrane to target human cells), which have more than a decade of safety data without a single death recorded.

As around 70 per cent of all new drugs being developed are genetically modified, experts say that it is essential that people understand that they will save lives. "If it is almost going to raise you from the dead and make you alive with a good quality of life, why would you not take it?" asks Shaw. "Every immunotherapy drug that you take for cancer treatment is a genetically modified antibody. Genetic modification has wonderful manifestations in terms of treatment options. These are amazing technologies. Usually, patients themselves don't have these concerns, it is the lay persons who don't understand and who are not in that predicament who make irresponsible comments."

The proliferation of chemotherapy is a good example of how to take new technologies to the masses. "Today, chemo drugs are made in India," says Dr Kumar, "which is why they are available in greater quantities and reduced cost. Involving communities such as ASHA workers also helped to get more people to stop fearing chemo. Trust is a big factor when you produce revolutionary new science."

And revolutionary the new science certainly is. If successful, immuno- and molecular oncology can completely alter India's cancer care landscape, extending a new lease of life to millions. For difficult cancers with low survival rates, the diagnosis will no longer be a death warrant. For other cancers, the earlier, simpler diagnosis will help both prevention and treatment. The affordability and accessibility of these new solutions in India may still be a long way off. But once here, the country's decades-long cancer burden could well become a chapter in ancient Indian history. ■